

SmartERP - Billing, Inventory & Accounting Management System

Python Enterprise Edition (Flask/FastAPI Version)

MVP FEATURES TO FOCUS ON :

LEDGER - SUPPLIER
CUSTOMER
ITEM / STOCK

VOUCHERS - SALES / CUSTOMER BILL
PURCHASE / INDIRECT STOCK ENTRY

1. Introduction

SmartERP is a cloud-based Billing, Inventory, Accounting, GST, and Business Management Platform inspired by TallyPrime and developed using modern Python technologies.

The system is designed to help businesses manage their daily accounting operations, inventory tracking, invoicing, GST reporting, customer management, supplier management, and financial reporting through a web-based interface.

One of the key goals of SmartERP is to replicate Tally's keyboard-first workflow, allowing accountants and operators to efficiently perform all operations without requiring a mouse.

The platform supports:

- Multi-company accounting
- Inventory management
- Billing & invoicing
- GST compliance - Dont Pay Attention for Project
- Financial reporting
- Banking operations
- Keyboard-only navigation

- Cloud deployment
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2. Project Objectives

The objective of SmartERP is to provide a complete business management solution that enables organizations to:

- Manage accounting records
 - Track inventory
 - Generate invoices
 - Monitor purchases and sales
 - Handle GST calculations
 - Produce financial reports
 - Improve business efficiency
 - Operate entirely using keyboard shortcuts
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3. Industry Relevance

ERP and accounting systems are essential for:

- Retail Stores
- Medical Shops
- Manufacturing Units
- Distributors
- Wholesalers
- Service Providers
- Chartered Accountant Firms
- Small & Medium Businesses

Applications such as:

- TallyPrime
- Zoho Books
- Busy Accounting
- QuickBooks

are widely used in industry.

Developing SmartERP demonstrates:

- Enterprise Application Development
- Python Backend Development
- Database Design
- Financial Systems Understanding
- Inventory Management
- ERP Architecture
- Business Workflow Automation

This type of project closely resembles real-world business software and provides significantly greater resume value than standard CRUD applications.

4. Technology Stack

Frontend

- Next.js
 - React.js
 - Tailwind CSS
 - ShadCN UI
 - TanStack Table
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Backend (Recommended)

Option 1 - Flask

- Python 3.12+
- Flask
- Flask-JWT-Extended / Spabase / Clerk / Appropriate / Convex
- SQLAlchemy / Supabase / PostgreSQL
- Alembic

Option 2 - FastAPI

- Python 3.12+
- FastAPI
- Pydantic

- SQLAlchemy / Supabase / PostgreSQL
 - JWT Authentication / Spabase / Clerk / Appropriate / Convex
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Database

- PostgreSQL / Supabase / PostgreSQL
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ORM

- SQLAlchemy
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PDF Generation

- ReportLab
-

Excel Export

- OpenPyXL
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API Documentation

- Swagger UI (FastAPI Built-In)
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Deployment

- Docker
- Render
- Railway
- Azure
- AWS

5. Application Workflow

Step 1 - Authentication

Users authenticate using:

- Email
- Password

Authentication handled through:

- JWT Tokens

After successful login:

User is redirected to Company Selection.

Step 2 - Company Selection

Each user account can manage up to 5 companies.

Available Actions:

- Create Company
- Select Company
- Alter Company
- Delete Company

Company Details:

- Company Name
 - Address
 - GST Number
 - Financial Year
 - Contact Information
 - State
-

Step 3 - Gateway of SmartERP

After selecting a company, the user enters the main dashboard.

Main Modules:

- Masters
- Transactions
- Inventory
- Accounting
- Banking
- GST
- Reports
- Administration

The workflow follows Tally-style navigation.

6. Masters Module

Ledger Management

Create and manage:

- Customer Ledger
- Supplier Ledger
- Expense Ledger
- Income Ledger
- Bank Ledger
- Cash Ledger

Operations:

- Create
 - Alter
 - Delete
 - Search
-

Group Management

Examples:

- Assets
 - Liabilities
 - Income
 - Expenses
-

Stock Groups

Examples:

- Electronics
 - Furniture
 - Grocery
 - Medical
-

Stock Items

Store:

- Item Name
 - SKU
 - HSN Code
 - Purchase Price
 - Selling Price
 - Quantity
 - GST Percentage
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Unit Management

Examples:

- PCS
 - BOX
 - KG
 - LTR
-

7. Voucher Management

Contra Voucher

Used for:

- Cash to Bank
 - Bank to Cash
 - Bank Transfers
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Payment Voucher

Examples:

- Rent
 - Electricity
 - Salaries
 - Vendor Payments
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Receipt Voucher

Used for:

- Customer Payments
 - Cash Receipts
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Journal Voucher

Used for:

- Adjustments
 - Corrections
 - Ledger Transfers
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Purchase Voucher

Functions:

- Add Purchase
 - Update Stock
 - Update Supplier Ledger
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Sales Voucher

Functions:

- Create Sale
 - Generate Invoice
 - Reduce Inventory
 - Update Customer Ledger
-

Credit Note

Sales Returns

Debit Note

Purchase Returns

8. Inventory Management

Features:

- Stock In
- Stock Out
- Stock Adjustment
- Stock Transfer
- Inventory Valuation

Track:

- Current Stock
 - Reserved Stock
 - Damaged Stock
 - Reorder Levels
-

9. Billing Module

Invoice Types:

- GST Invoice
- Proforma Invoice
- Estimate
- Quotation

Features:

- Auto Invoice Number
 - Tax Calculation
 - PDF Download
 - Print Invoice
 - Email Invoice
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10. Customer Management

Store:

- Customer Name
- GST Number
- Mobile Number
- Address
- Outstanding Balance

Functions:

- Add Customer
- Edit Customer
- View Statements

- Ledger View
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11. Supplier Management

Store:

- Supplier Details
 - Purchase History
 - Payment History
 - Outstanding Dues
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12. Banking Module

Features:

- Bank Ledger
 - Fund Transfers
 - Bank Reconciliation
 - Cheque Management
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13. GST Module

Tax Types:

- CGST
- SGST
- IGST

Features:

- GST Calculation
- GST Register
- Tax Reports
- Tax Summary

14. Reports Module

Accounting Reports

- Balance Sheet
- Profit & Loss
- Trial Balance
- Cash Flow

Inventory Reports

- Stock Summary
- Low Stock Report
- Item Movement Report

Sales Reports

- Daily Sales
- Monthly Sales
- Customer Sales

Purchase Reports

- Purchase Register
- Supplier Summary

GST Reports

- GST Summary
- Tax Collection

15. Keyboard-Only Navigation System

The entire application should be operable without a mouse.

Global Shortcuts

F1 = Company Selection

F2 = Change Financial Year

F3 = Company Information

F4 = Calculator

ESC = Previous Screen

CTRL + H = Dashboard

CTRL + K = Global Search

CTRL + Q = Logout

Masters

ALT + L = Create Ledger

ALT + A = Alter Ledger

ALT + G = Create Group

ALT + S = Create Stock Item

ALT + U = Create Unit

Voucher Shortcuts

F5 = Payment Voucher

F6 = Receipt Voucher

F7 = Journal Voucher

F8 = Sales Voucher

F9 = Purchase Voucher

ALT + F8 = Credit Note

ALT + F9 = Debit Note

Inventory

CTRL + I = Inventory Dashboard

CTRL + N = New Item

CTRL + E = Edit Item

CTRL + T = Stock Transfer

CTRL + R = Stock Report

Billing

CTRL + B = Create Invoice

CTRL + P = Print Invoice

CTRL + SHIFT + P = PDF Export

CTRL + M = Email Invoice

Reports

ALT + B = Balance Sheet

ALT + P = Profit & Loss

ALT + T = Trial Balance

ALT + C = Cash Flow

ALT + R = Stock Summary

ALT + X = GST Reports

16. Database Design

Core Tables:

- users
 - companies
 - groups
 - ledgers
 - stock_groups
 - stock_items
 - units
 - customers
 - suppliers
 - purchases
 - sales
 - invoices
 - voucher_entries
 - inventory_transactions
 - gst_records
 - audit_logs
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17. Recommended Backend Architecture

backend/

|— app.py
|— routes/
|— controllers/
|— services/
|— models/
|— schemas/
|— middleware/
|— auth/
|— inventory/
|— accounting/
|— gst/
|— reports/
|— utils/
|— database/
└— config/

18. Two-Week Development Schedule

Week 1

Day 1

- Requirement Analysis
- Database Design

Day 2

- Flask/FastAPI Setup
- PostgreSQL Setup

Day 3

- Authentication Module

Day 4

- Company Management

Day 5

- Dashboard Development

Day 6

- Ledger Management

Day 7

- Group Management
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Week 2

Day 8

- Inventory Module

Day 9

- Purchase Voucher

Day 10

- Sales Voucher

Day 11

- Billing System

Day 12

- Reports Module

Day 13

- Keyboard Shortcut Integration

Day 14

- Testing & Deployment
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19. Resources

Python

<https://docs.python.org/3/>

Flask

<https://flask.palletsprojects.com/>

FastAPI

<https://fastapi.tiangolo.com/>

SQLAlchemy

<https://docs.sqlalchemy.org/>

PostgreSQL

<https://www.postgresql.org/docs/>

ReportLab

<https://www.reportlab.com/documentation/>

OpenPyXL

<https://openpyxl.readthedocs.io/>

YouTube Channels

- freeCodeCamp
 - Tech With Tim
 - Corey Schafer
 - Traversy Media
 - Hitesh Choudhary
 - CodeWithHarry
-

GitHub References

<https://github.com/topics/erp>

<https://github.com/topics/accounting-software>

<https://github.com/topics/flask>

<https://github.com/topics/fastapi>

<https://github.com/topics/inventory-management>

Tally Tutorial

 Tally Class - Lecture 1 | टैली कैसे सीखें | Tally ERP 9 Course in Hindi

20. Future Enhancements

- Barcode Scanner

- Multi-Branch Management
 - Payroll Module
 - Role-Based Permissions
 - WhatsApp Invoice Sharing
 - OCR Bill Scanning
 - AI Business Insights
 - Mobile Application
 - E-Invoice Integration
 - UPI Reconciliation
 - Bank API Integration
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21. Conclusion

SmartERP is a modern Tally-inspired ERP platform built using Python technologies that combines Billing, Inventory Management, Accounting, GST, Banking, Reporting, and Business Analytics into a single web application.

The project demonstrates:

- Python Backend Development
- ERP System Design
- Financial Workflow Automation
- Database Architecture
- Security Implementation
- Inventory Management
- Reporting Systems
- Enterprise Software Development

This project is highly valuable for Full Stack Python Developers, Backend Developers, and Software Engineers because it closely resembles real-world ERP systems used by businesses across multiple industries.

A completed SmartERP project can serve as a flagship portfolio project and significantly strengthen a resume by showcasing advanced business application development skills.